

Nathan Liu

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EDUCATION

University of California, Berkeley

Berkeley, CA

Bachelor of Arts in Data Science (Applied Math) & Computer Science

August 2026

TECHNICAL SKILLS

Languages: Python, Java, SQL (PostgreSQL), C/C++, TypeScript

Developer Tools: Git, Docker, Kubernetes, AWS (DynamoDB, S3, RDS, EC2)

Frameworks: Django, FastAPI, React, Node.js, PySpark

Libraries: pandas, NumPy, Matplotlib, PyTorch

PROFESSIONAL EXPERIENCE

Data Engineer Intern

May 2025 – August 2025

Carbon Sustain

Berkeley, CA

- Engineered a real-time event-processing pipeline handling 100K+ events/day, adding schema validation and S3 execution traces to support deterministic replay and faster incident debugging.
- Contributed to enhancing system reliability by implementing DynamoDB idempotency keys to handle retry spikes and configuring CloudWatch alarms and cutting real-time execution failures.
- Analyzed catering and digital-media footprint data to identify decarbonization opportunities representing up to 76% of modeled event emissions.

Software Engineer

January 2025 – May 2025

UnifIBD

Remote (San Jose, CA)

- Streamlined analytics-to-production workflows by standardizing feature integration and engineering handoffs, cutting delivery time by 50%.
- Removed manual mobile releases to TestFlight and Google Play with Fastlane and GitHub Actions, reducing deployment time from hours to minutes.
- Led a three-engineer migration of a Swift iOS application to Flutter, defining the architecture and technical milestones while consolidating platform-specific code into a shared codebase.

Data Science Intern

June 2024 – August 2024

L.A. Lucky Import & Export Inc.

Los Angeles, CA

- Developed statistical forecasting pipelines and inventory anomaly detection across 4,000+ SKUs, identifying seasonal surges to reduce inventory shrinkage by 22%.
- Reduced API response latency by 55% by minimizing response payloads and introducing server-side caching for frequently accessed inventory data.
- Reconstructed a real-time SAP inventory sync using incremental pulls and upserts into PostgreSQL which reduced manual processing substantially and enabled automated invoicing for all customers.

PROJECTS

CardboardDex | TypeScript, Python, PostgreSQL

2026

- Built a production TCG price tracker full pipeline from eBay listings through a custom NLP title matching, served via FastAPI + Next.js on a hybrid Cloudflare + AWS stack with automated CI/CD.
- Engineered high-performance search over 66,000+ cards using multi-tier caching achieving sub-5ms queries against a live AWS RDS instance tracking 280,000+ price observations.

FleetManager | Python, Rust, PostgreSQL, AWS

2025

- Developed a risk-scoring pipeline combining driver behavior and vehicle-utilization signals to prioritize vehicles for preventive maintenance.
- Built Django/PostgreSQL REST APIs for bulk fleet telemetry ingestion, AWS S3 storage, utility-boundary resolution, and predictive trip modeling.